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An Adaptive Preview Path Tracker for Off-Road Autonomous Driving

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Abstract— **A path tracking algorithm with adaptive preview distance based on roadside sampling is proposed in order to solve the path tracking control problems of autonomous driving in the complex off-road environment. The proposed lateral and longitudinal control laws involve both feedback control and feedforward control. We have reformulated the lateral control problem as an optimal control problem, and the designed trajectory is planned by taking the kinematic constrains of the vehicle and the preview road information pertaining to road curvature and road width into account. The desired lateral control command is directly decided by the curvature of the planned circular curves and the longitudinal velocity is planned by utilizing the preview information and the planned circular curves. In the meantime, the feedback control laws are designed by adopting inverse model of the vehicle dynamics. We implement our control method on the 14-DOF vehicle simulation model. And the simulation results are presented and discussed.**

I. INTRODUCTION

A. Background

The past three decades have witnessed an increased research effort in the area of autonomous driving. As one of typical mobile robots, autonomous vehicle has received a lot of research interests in robotics community. There have been large amount of research and applied contributions of navigation control technologies for autonomous vehicles in the structured environments, while in the complex off-road environments there are still numbers of challenging problems to be faced. As one of the active areas of off-road autonomous vehicle research field, path tracking control has always been studied. It plays a critical role for the safely, stably and rapidly executing tasks for off-road autonomous vehicles.

B. Related work

Path tracking refers to following a desired geometric path of path planning system by applying appropriate steering motion and speed control commands to the vehicle. The goal of typical path tracking control strategy is to minimize the lateral distance error and heading error between the vehicle and defined path while maintaining the stability [1]. In complex off-road environments, the curvature, road-width, terrain conditions of the road always changed over a wide range, which have adverse effects on the vehicle's control system. In order to achieve high performance, the path tracking control system should be able to take the environment constrains, vehicle kinematics and dynamic constraints into account [2].

Researchers of autonomous vehicle have done a large amount of research work in order to solve the path tracking problem for the Ackerman steered driving vehicles in the complex off-road environment [3]. Early research work of autonomous vehicle had formulated the path tracking problem as a regulation control problem. The control law was simple, which utilized the simplified dynamic model of vehicle and merely reacted to the current lateral and heading errors. Many variants of the feedback control strategy have been proposed for modern applications [4]. For instance, several lateral control algorithms have been applied in commercial vehicles toward safe and high-speed operation, e.g. (ESP (Electric Stability Program), ESC (Electronic Stability Control)) and VDC (Vehicle Dynamics Control)) [5]. Some of the feedback control approaches have achieved high tracking performance for autonomous driving in structured environments, whereas they will quickly reach their limits and results in poor performance or even leads to the instability of integral control systems in off-road environment. The famous Standley algorithm used a nonlinear control law for off-road autonomous driving, which took advantage of current heading errors and lateral errors. The global asymptotical stability was proved as well [3]. However, the stability of the path tracking control was depended on the smoothness of the predefined path. In order to smooth the prior path planning results, Dolgov *et al*. [6] presented a complicate conjugate gradient approach. Most of the feedback methods referred above have a common shortage of control parameter adjustment. In addition, the stability of the control system has been largely depended on smoothness of the predefined path.

We find that most of these the feedback control strategies merely react to and try to compensate the states' errors and neglect preview information. As a matter of fact, the curvatures, road-width and the roughness of terrain in off-road environment may vary over a large range, so the gains of the feedback control methods may be adjusted frequently to compensate the state errors. Thus it may easily result in instability of the path tracking control system. However, many of research on human driver models have shown that human drivers typically take advantage of the finite-horizon preview road information ahead, predict the desired response trajectories, and steer the vehicle to minimize path errors in an optimal fashion [7] [8] [9]. The finite-horizon preview

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information of the road is able to be sensed through the on-board perception systems, and it could be used for path achieved high performance. It had explicitly improved the robustness and adaptive capacity of the tracking control system [7].

One of the most widely applied path tracking algorithms based on preview information is pure pursuit method. Because of its simplicity and robustness, it has been widely implemented even on some well known autonomous vehicle projects [10]. The pure pursuit method computes a circular curve in real time, which connects the center of the rear axle and the preview goal on the desired path ahead of the vehicle [11]. The algorithm implicitly meets the nonholonomic constraints and utilizes the information related to the upcoming road characteristics. Though only satisfying the position constraints, the algorithm exhibits to be adapted to the varying curvatures of road. However, the performance of the algorithm is closely depended on the look-ahead distance.

In order to generate the feedforward commands for lateral control and longitudinal control by utilizing the information related to the desired path ahead and meeting the vehicle kinodynamic constraints, several approaches are presented. Howard *et al*. [12] reformulates the problem as an optimal control problem. Firstly the algorithm determines a set of sub-goal states along the predefined path, and a trajectory generation method is proposed in special parametric spaces to generate a set of trajectories which could meet the arbitrary state constraints of the starting and ending points [13]. Then the optimal trajectory is determined dynamically by minimizing a utility function, which includes the factors that affected the control performance. The trajectory generation algorithm also considers the wheel-terrain interaction effects on the vehicle dynamic model. However, the convergence could not be guaranteed. Werling *et al.* [14] deal with the trajectory generation problem through an alternative way. The lateral and longitudinal profiles are generated by using quintic polynomials with time parameter in order to ensure the continuous acceleration, while the method is applied in on-road environments based on the assumption that the curvature of the desired path is known.

To cope with the problem of adaptive look-ahead distance and enhance the path tracking performance of pure pursuit algorithm, a real-time path tracking algorithm base on roadside sampling is proposed in this paper. In addition, the longitudinal velocity planning is also taken into account.

The paper is organized as follows. The problem of typical pure pursuit is described in Section II. An adaptive preview distance path tracking algorithms is presented in Section III, and the lateral control law base on inverse model of vehicle is proposed in Seciton IV. Section V shows the simulation results. Finally the conclusions are given in Section VI.

II. PROBLEM STATEMENT

Based on the related work, the proposed path tracking control framework consists of both preview controller and feedback controller.

tracking control. Many researches had combined the feedback control and preview control for path tracking purpose and

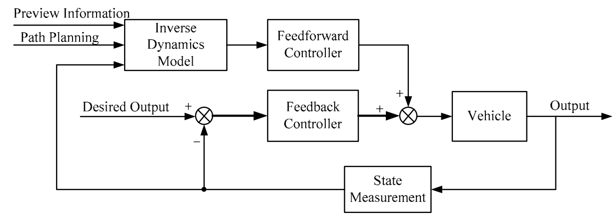


Figure 1. Framework of path tracking control system.

As shown in Figure 1 the feedforward control is calculated by combining the finite-horizon preview road information with the path planning and state measurement information of the vehicle. In order to solve the problem of look-ahead distance decision, an algorithm based on roadside sampling is presented, which considers both the road width and road curvature ahead of the vehicle. To a large extent, the driving commands are decided by the feedforward information. It responds to the curvature of the upcoming road. The lateral control commands should be smoothed in order to minimize the negative effects of tracking the curvy road. Most importantly, vehicle kinematic and dynamic constraints should be taken into account as well. In addition, the longitudinal velocity is planned in order to improve the stability and efficiency of autonomous driving.

A. Geometric model of the Ackerman steered vehicle

The path tracking algorithm proposed in the paper is mainly implemented on the Ackerman steered vehicles. Firstly the geometrical characteristics of the typical vehicle should be taken into account, which implies the kinematic constraints of the vehicle. It is a common way to simplify the Ackerman steered vehicle as a bicycle model, which combined the two front wheels to form the front wheel and two rear wheels to form the rear wheel respectively.

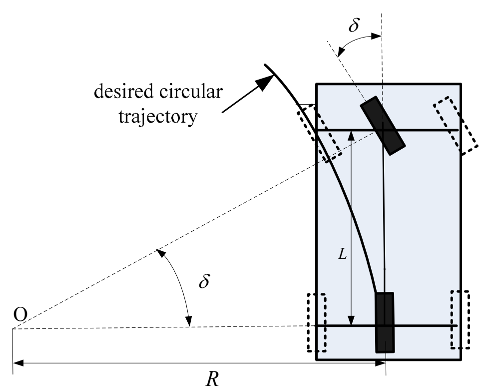


Figure 2. Equivalent bicycle model of Ackerman steered vehicle.

Figure 2 indicates the approximate steady relationship between the steering angle of the front wheel and the curvature of the circular path that the rear wheel will execute. Thus, the geometric relationship can be written as:

$$\tan\delta = \frac{L}{R} = Lk \tag{1}$$

Where δ denotes the steering angle of the front wheel, L denotes the wheelbase, R denotes the radius of the desired following circular path and k is the curvature equals $1/R$.

The geometrical model will approximate the motion of the Ackerman steered vehicle at low speed and moving on a plane. The ideal model neglects the dynamic effects, e.g. the cases of oversteer and understeer will happen when the vehicle is running on the varing terrain at high speed. Besides, the velocity of the car has an unavoidable effect on the lateral dynamic model. Accordingly in order to compensate the disturbances, the feedback control law should be designed deliberately.

B. *Typical pure pursuit strategy*

According to the geometric characteristics of the Ackerman steered car aforementioned, the typical pure pursuit strategy assumes that the desired trajectory of the vehicle could be approximate by circular arcs. As shown in Figure 3, a preview point is determined on the predefined path ahead, and the distance between the rear wheel and the preview point is l_d. In order to follow the expected preview point, a circular arc is generated by connecting the center of the rear axle and the preview point [1]. According to the geometrical relationships, the lateral control law could be attained as

$$k_d = \frac{2e_d}{l_d^2} \tag{2}$$

The desired trajectory is approximate by circular arc, where k_d is the curvature of the trajectory. The e_d is the lateral error between the center of the real wheel and the preview point.

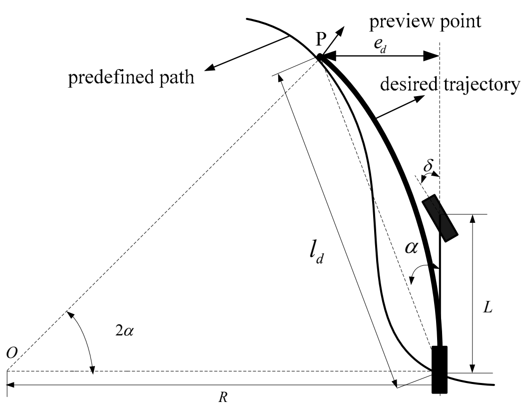


Figure 3. Typical pure pursuit tracking strategy.

Then the geometric relationship as Equation (1) is exploited to generate the lateral control law to track the generated circular trajectory. Here we find that lateral control law is closely dependent on the preview distance. Commonly the preview distance is determined to be proportional to the longitudinal velocity and saturated at a minimum and maximum value. However, the fixed preview distance could not be adapted to the varying curvatures and the varying widths of the road. A short look-ahead distance provides more accurate tracking, but it will cause instability when the road is too curvy. On the contrary, a longer look-ahead provides smoother tracking and explicitly improves the stability of lateral control, but it will result in “cutting corners” when executing a sharp turn.

The typical pure pursuit algorithm supposes that the desired path is a curve. The vehicle tracked the desired path without considering the road width. In on-road environments, it is reasonable, e.g. in the express way environment, the designed path could be the lane and the lane width is not varied over a large range. Besides, the curvature of the lane could be approximately considered to be continuous. Whereas in the off-road environments, the road width and curvatures varies at a large range width, commonly it’s very difficult to attain a smooth desired path by the path planning system. It’s not reasonable for the path tracking control system to track the desired path without considering the road width. Besides the velocity plan should be considered as well.

Above all, the road environment should be combined with the predefined path to determine the preview distance and the preview goal point.

III. Strategy of Feedforward Control

A. *Description of the road in the vehicle coordinate*

Considering the effects of the road width, the collision avoidance is tested by sampling right and left roadsides along the predefined path. The road width is saturated at a maximum value to limit the heading of the path and a minimum value to keep off the roadsides. Besides, the vehicle width should also be taken into account. Thus, a road with right and left roadsides was obtained.

Firstly the local road environment maps the distributions of the road and obstacles to body-frame coordinate system. A body-frame coordinate system is defined with the positive x-axis pointing forward, the positive y-axis pointing to the left. The position of the center of the rear wheel is $(x_0, 0)$, heading angle is 0. The predefined path by the path planning system is described as the curve $f(x)$. The discretization formation of the curve is described as $(x_i, f(x_i))$ $(i = 0,1,2,\ldots N)$, as shown in Figure 4.

The collision test results in the left roadside $f_l(x_i)$ and the right roadside $f_r(x_i)$. We can also attain the center line of road $f_m(x_i)$, and $f_m(x_i)=(f_l(x_i) + f_r(x_i))/2$.

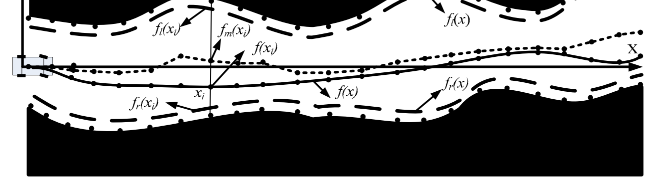


Figure 4. The description of the local road environments in the body-frame coordinate.

B. The algorithm of determining the feedforward control based on roadside sampling

Typical pure pursuit algorithm shows that the desired trajectory is determined by the preview distance. Here we divide feedforward control strategy into two sequential steps, trajectory planning and trajectory optimization. In the first step, we sample the roadsides $f_l(x)$ and $f_r(x)$ instead of predefined path $f(x)$ to determine the feasible trajectory sets. Then in the second step the optimized trajectory is determined by minimizing an objective function based on the lateral error, preview distance and smoothness of the control output.

First of all, the parameters should be initialized. The initial preview distance $d_p = x_1$; the maximal distance of the local environment along the x-axis is l_{max}; the discretization step is l_{step}; the total sampling points N = l_{max}/l_{step}; the current curvature of the vehicle is k_0; Considering mechanical constraints of the vehicle steering angle, the desired curvature $|k| <= k_{max}$; initialize the maximal left curvature the circular arcs $k_{lmin} = k_{max}$, and initialize the minimal right curvature of the circular arcs $k_{rmax} = -k_{max}$.

In order to find all of the feasible trajectory sets, we sample the left and right roadside point by point as Algorithm 1 shows [15]. Connection function returns the curvature of the circular arc.

Algorithm 1 Find_Feasible_Trajectories()

Input: $x_i, f_l(x_i), f_r(x_i), f_m(x_i), i=1,2,\ldots,N$
Output: i_{max}, d_p, $s(x_i)$, $k(x_i)$
for $i = 1$ **to** N **do**
  $k_l(x_i) \leftarrow$ Connection $(x_0, x_i, f_l(x_i))$;
  $k_r(x_i) \leftarrow$ Connection $(x_0, x_i, f_r(x_i))$;
  $k(x_i) \leftarrow$ Connection $(x_0, x_i, f_m(x_i))$;
  $s(x_i) \leftarrow$ Arc_Length $(x_0, x_i, f_m(x_i))$; //calculate the arc length
   if $|k(x_i)| > k_{max}$ **then**
    continue;
   else
    if $k_l(x_i) < k_{lmin}$ **then**
     $k_{lmin} \leftarrow k_l(x_i)$;
    if $k_r(x_i) > k_{rmax}$ **then**
     $k_{rmax} \leftarrow k_r(x_i)$;
    if $k_{lmin} < k_{rmax}$ **then**
     $i_{max} \leftarrow i$-1;
     break;
  $d_p \leftarrow x_i$;
  return i_{max}, d_p, $k(x_i)$, $s(x_i)$;

After all of the feasible trajectories are generated, we try to find the optimal trajectory by minimizing the utility function. The preview distance, the cross error and the smoothness of the trajectory are considered in the utility function. The utility function is written as

$$i^* = arg \min_{i=1}^{i_{\max}} \{\omega_1 \frac{x_N}{x_N + x_i} + \omega_2 \sum_{m=1}^{i} \frac{\left\| f_d(x_m) - f_m(x_m) \right\|_2}{s(x_i)} + \omega_3 \left\| k(x_i) - k_0 \right\|_2 \} \quad (3)$$

Where i^* is the index of the optimal trajectory. ω_1, ω_2, ω_3 are the weighted coefficients. $f_d(x)$ is the y-axis values of desired trajectory. Finally we obtain the optimal trajectory which connects the current position to the optimal desired preview point (x_{i^*} , $f_m(x_{i^*})$).

We attain the feasible steering control commands space by sampling the roadsides and ensure the free-collision efficiently. For the optimal trajectory generation algorithm considering the factors of the road width, road curvature preview, the look-ahead distance is determined to be adapted to the dynamically changes of the upcoming road environments. Besides, the lateral error and the smoothness of the steering command are also taken into account to ensure the stability of the lateral control.

IV. Strategy of Feedback Control

For the purpose of tracking the desired trajectory accurately and rapidly, the lateral and longitudinal feedback controller is very necessary.

A. The lateral feedback controller

The lateral feedforward control law utilizes the geometric steering characteristics shown in Equation (1). However, the control law neglects the side slip angle of the wheel and the open-loop control strategy will lead to large error when the vehicle moves on a rough terrain at a high speed. It may also result in oversteer or understeer. In order to eliminate the external disturbances, we utilize the inverse dynamic model of the vehicle to design the internal model control structure as Figure 5 shows.

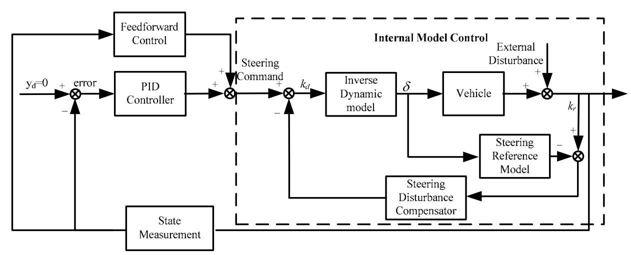


Figure 5. Framework of lateral control.

As we know that it's too complicate to create the accurate nonlinear dynamic models of the car. Besides, many of states of the model are not available by the on-board sensors. Here we use the steady steering characters to establish the nominal model of the steering control system. The steady steering angle is shown as Equation (4).

$$\delta = Lk + (\frac{l_r}{C_{\alpha f}} - \frac{l_f}{C_{\alpha r}}) \frac{mv_x^2}{L} k \quad (4)$$

Where m denotes the mass of the vehicle; l_r denotes the longitudinal distance from c.g. of vehicle to the rear wheel; l_f denotes the longitudinal distance from c.g. of vehicle to the front wheel; $C_{\alpha f}$ and $C_{\alpha r}$ denote the cornering stiffness of the front and rear wheels respectively.

By utilizing the internal model of the vehicle lateral control, the curvature of the desired trajectory at a certain steering angle can be predicted. We can take advantage of the error between the desired curvature and the real curvature of the vehicle to reject the lateral disturbances as well. In addition, the response speed and stability of the lateral control system could be improved too.

In order to avoid the sideslip, the maximal lateral acceleration a_{max} should satisfy constrain as follows:

$$|k_d| \le \frac{a_{\max}}{v_x^2} \tag{5}$$

B. The longitudinal velocity planning

It is common that the autonomous vehicle research community addresses the path tracking control issue with only considering steering motion control. In fact, the longitudinal velocity planning is significant for the autonomous driving in off-road environments as well [16].

Firstly the maximal longitudinal velocity v_{max} should be limited according to the maximal lateral acceleration, the proximity to the roadside and velocity limits of the road conditions.

After the maximal longitudinal velocity limit is attained, we plan a trapezoidal profile for longitudinal velocity as Figure 6 shows. In order to guarantee the safety of autonomous driving, the terminal value of the trajectory is often set to be zero. Due to the delay of the servo system, the minimal safety distance s_{safe} also should be considered, and the integral of the velocity from t_0 to $t_{terminal}$ is s-s_{safe}. Where s denotes arc length of the planning trajectory.

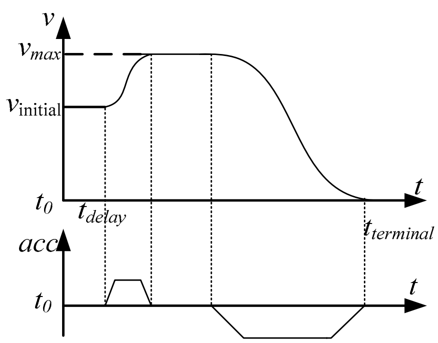


Figure 6. Framework of lateral control.

In the real time operation, the abrupt switch of the acceleration should be avoided because of the delay of the propulsion system of vehicle. In order to improve the control performance of longitudinal control, the spline fitting method is utilized to smooth the velocity profile [17].

V. Experimental Results

We establish the simulation model by using Matlab tools. The proposed path tracking control algorithms is implemented on the 14-DOF vehicle dynamic model, which is developed to precisely represent the vehicle dynamic response [18]. The dynamic model describes the traction and braking systems, the vehicle body and suspension systems and the unified tire models. We apply the path tracking algorithm on different experiments to evaluate the control performance. The planning and control cycle time of the control system is 100ms and 50ms respectively. The maximal and minimal preview distances are 30m and 5m respectively. The main parameters of the vehicle in the simulation experiments are listed as follows:

TABLE I. Main Parameters of Vehicle Dyanmic Model

Paramters	Symbol	Value
mass (kg)	m	2160
moment of inertial (kg·m^2)	I_z	3280
conering stiffness (N/rad)	$C\alpha_f$, $C\alpha_r$	56000, 66000
vehicle width (m)	d	1.5
base length (m)	l_f, l_r	1.5, 1.3

In order to avoid the collision with obstacles, the minimal keep-off distance is set 0.7m. The states of the vehicle are attained by the numerical integrated of the dynamic model.

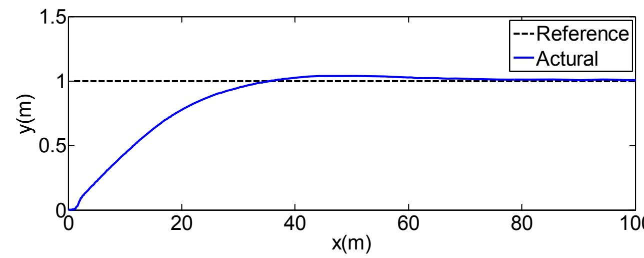
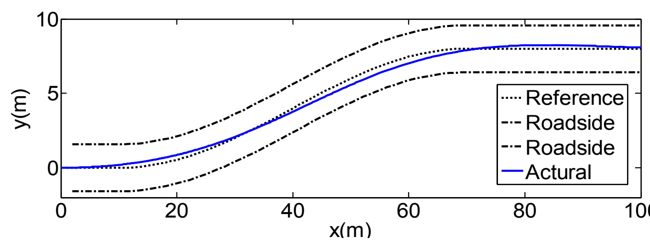
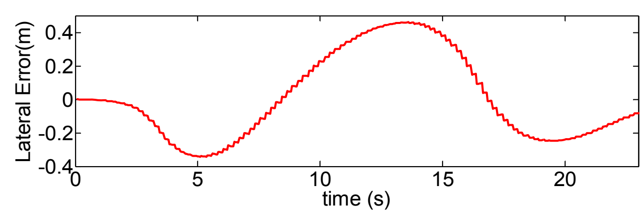


Figure 7. Tracking control of steady lateral error.

Figure 7 shows the tracking response of the system when the steady lateral error exists. The path tracker is able to compensate the lateral error and stabilize the control system, though overshoot exists.



(a)



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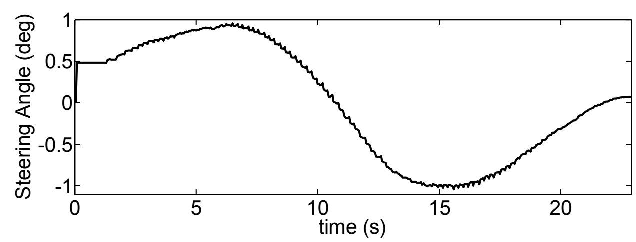
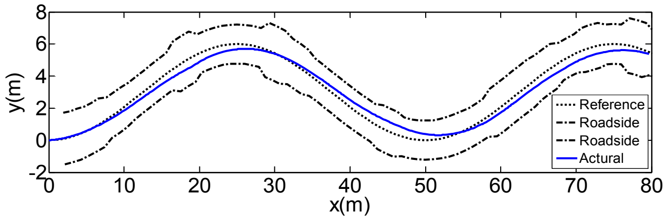


(c)

Figure 8. Tracking control of continous varing curvature,with 4m fixed road width. (a) Trajectotry. (b) Lateral error. (c)Steering Angle.

Figure 8 displays the results of tracking continuous target path with fixed road width. The vehicle could track the target path smoothly, though lateral error exists. The steering command is smoothed to reject the abrupt action.



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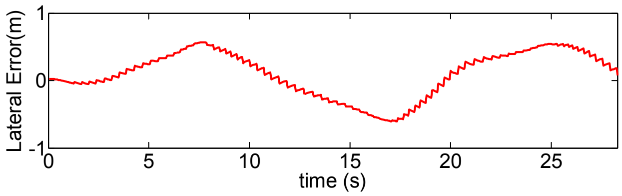

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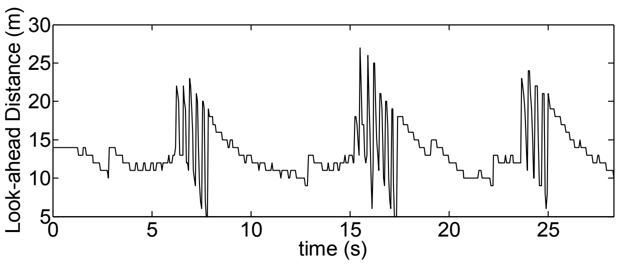

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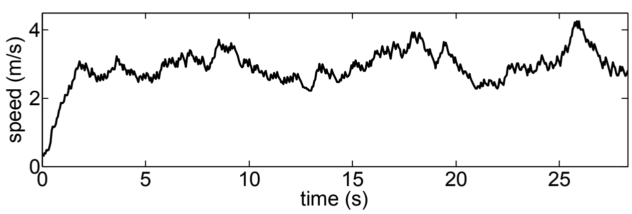

(d)

Figure 9. Tracking control of path with varing curvature, and varing road width. (a) Trajecotry. (b) Lateral Error. (c) Look-ahead Distance. (d) Speed.

Figure 9 shows the experimental results of tracking the path with discontinuous curvature and varying road width. Even though the path is curvy, the look-ahead distance can be adjusted to be adapted to the curvy road. Accordingly, the longitudinal velocity is also varied with the varying of the look-ahead distance.

VI. Conclusions

In this paper, we present an adaptive path tracking algorithm with feedforward control and feedback control in order to address the path tracking problem for autonomous driving in the off-road environment. The feedforward control command is determined based on the optimization of a utility function which deliberately takes the varying curvatures, the road width and the smoothness of the control commands into account. The feedback controller and longitudinal velocity planner is designed for the sake of tracking the desired path accurately and rapidly. The simulation results have validated the feasibility and effectiveness.

For future work, more experiments need be implemented on the real autonomous vehicle platform to verify and enhance the performance of the proposed path tracking algorithms.

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